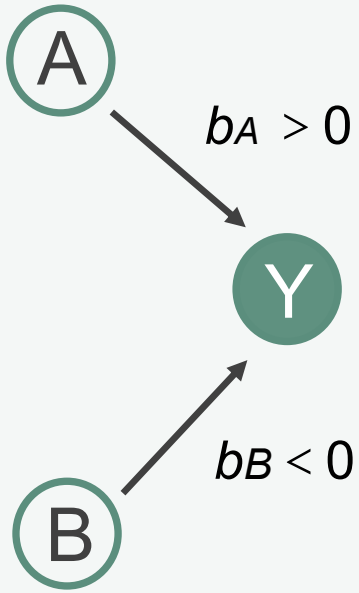
$R = A/B \longrightarrow \log(R) = \log(A) - \log(B)$

Ratio-imposed constraint
$b_A + b_B = 0$

A

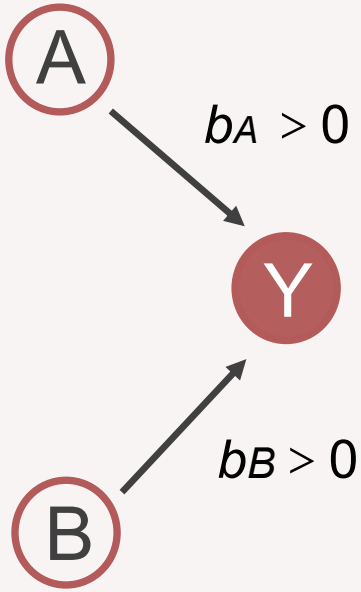
Constraint-compatible effects



Opposite signs
Equal magnitude

B

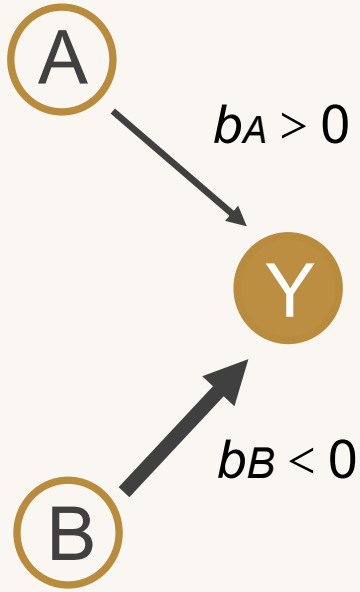
Same-direction effects



Same sign
Constraint violated

C

Magnitude imbalance



Opposite signs
Constraint violated